

Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
University of Pisa

Project Report

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**Project Title: Predicting Online Shoppers purchasing
Intention**

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Abstract

This project addresses the challenge of predicting online shoppers' purchasing intentions to optimize e-commerce purchasing possibilities. Using the "Online Shoppers Purchasing Intention" dataset, the problem was framed as a binary classification task to predict whether a browsing session will end in a transaction. The analysis revealed a class imbalance, which was addressed using the SMOTENC oversampling technique. To ensure a robust model evaluation, a 10×10 Nested Cross-Validation strategy was implemented. The study compared five different algorithms, prioritizing F1-score (and specifically Recall) over global Accuracy, given the asymmetric costs of misclassification in e-commerce. The results indicate that CatBoost outperformed the other models. Sequentially, the winning model passed through a final hyperparameter tuning and was trained on the complete dataset. Initial feature importance analysis revealed severe indirect target leakage from opaque Google Analytics metrics. An ablation study was conducted by excluding these features, achieving a final F1-score of 0.37. Finally, Explainable AI techniques, specifically SHAP, were applied to the model to provide both global and local interpretability regarding the true drivers of online purchasing intent.

1. Introduction

In e-commerce, it is very important to know if a user wants to buy something. Web analytics tools collect a lot of data about user sessions, however only a few users make a purchase. If a business can predict who the possible buyers are, it can offer them special discounts or help to ensure the deal. To achieve this, different algorithms can be evaluated and compared to find the best predictions [1]. The main problem is that the data is highly imbalanced, there are many more non-buyers than buyers and missing a real buyer is much worse than showing a discount to a user who will not buy [2]. Finally, it is important to use Explainable AI (XAI) to understand how the model makes its decisions. During this part the model will be analysed, and the problem could be treated in different ways.

This project aims to:

1. Develop a predictive model for a binary classification on an imbalanced target variable.
2. Compare the performance of multiple classification models by using a Nested Cross-Validation and evaluating the impact of the SMOTENC technique.
3. Provide global and local model interpretability by extracting SHAP values from the winning model.

References

- [1] (Reza Alamsyah, Sri Wahyuni), (2024). "Using Random Forest and Support Vector Machine Algorithms to Predict Online Shopper Purchase Intention from E-Commerce Session Data." International Journal of Artificial Intelligence and Machine Learning.
- [2] Sakar, C. O., Polat, S. O., Katircioglu, M., & Kirmizi, Y. (2019). "Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks." Neural Computing and Applications, 31(10), 6893-6908.

2. Problem

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Binary classification
Dataset	UCI Online Shoppers Purchasing Intention Dataset— 12330 samples, 17 features
Target variable	Revenue (Boolean: True / False)
Features	17 features: 10 numerical, 7 categorical
Missing values	0 missing values
Class distribution	False: 84.5%, True: 15.5% — unbalanced
Evaluation metric(s)	F1-score and particularly recall
Train / Val / Test	10x10 Nested Cross-Validation

2.1 Exploratory Data Analysis

The dataset contains exactly 0 missing values across all 12,330 samples. Figure 1 shows the class imbalance of the target variable Revenue. Non-purchasing sessions represent 84.5% of the data, while successful transactions account for only 15.5%. Figure 2 shows the correlation heatmap of the numerical features. The analysis reveals high correlation between two feature pairs: BounceRates - ExitRates and ProductRelated - ProductRelated_Duration.

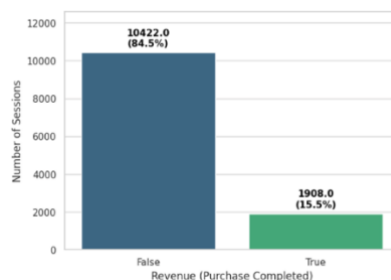


Figure 1. Distribution of the 'Revenue' variable

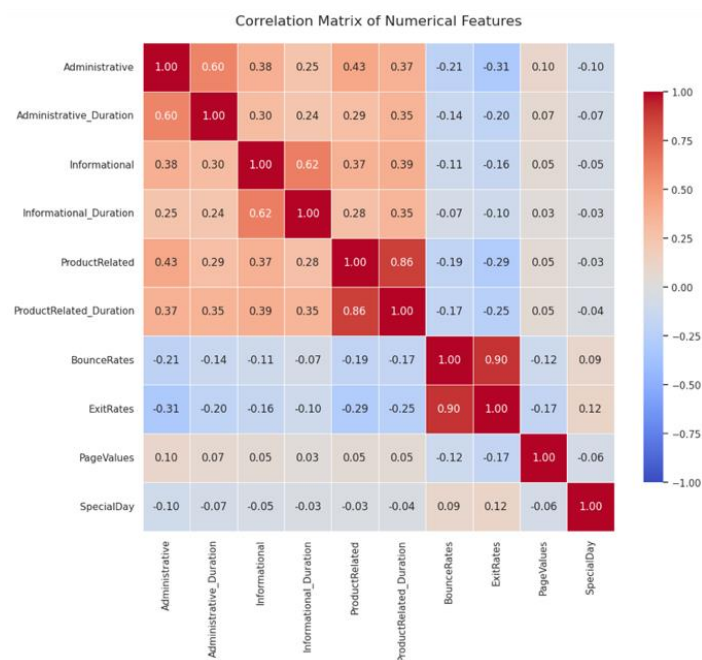


Figure 2. Correlation heatmap of the numerical features.

2.2 Feature Description

The dataset captures real-time web session data and user demographics. The original 17 features are categorized into groups:

- Administrative, Informational, ProductRelated (Numerical): Count of pages visited per category. The system derives these values from URL changes during the session.
 - Administrative_Duration, Informational_Duration, ProductRelated_Duration (Numerical): Total time spent in each page category, computed using session timestamps.
 - BounceRates, ExitRates, PageValues (Numerical): Metrics extracted from Google Analytics regarding average values of the pages visited.
 - SpecialDay (Numerical): Closeness to a holiday, ranging from 0 to 1. The authors computed this value based on order-to-delivery dynamics.
 - Month (Categorical) & Weekend (Boolean): Month and weekend status of the visit.
 - OperatingSystems, Browser, Region, TrafficType (Categorical): User identifiers. The system extracts these variables from HTTP requests and session cookies.
 - VisitorType (Categorical): Specify what type of user is browsing (New/Returning/Other).
 - Revenue (Boolean): The target variable of the task. It indicates whether the session successfully ended with a transaction (True) or not (False).
-

3. Pipeline and Experiments

3.1 Preprocessing

The preprocessing stage consists of data cleaning and in-fold transformations.

The raw dataset contained 125 duplicate sessions, these few records are removed to prevent data leakage. No missing values are found in the dataset; therefore, no imputation strategy is required.

Four variables, OperatingSystems, Browser, Region, and TrafficType, are stored as integer values. These variables represent nominal categories rather than ordered quantities, they are converted to string format to ensure correct treatment as categorical features. This conversion was also required by SMOTENC, which processes numerical and categorical variables separately.

The machine learning pipeline applies preprocessing transformations through a ColumnTransformer. Numerical features are standardized using StandardScaler improving numerical stability. Categorical features are encoded using OneHotEncoder.

All preprocessing steps are embedded inside the machine learning pipeline and are fitted exclusively on the training portion of each fold. Validation and test samples never influence scaling or encoding parameters. This design prevents data leakage.

3.2 Feature Selection and Engineering

Figure 2 shows two highly correlated feature pairs:

- BounceRates and ExitRates
- ProductRelated and ProductRelated_Duration

For each pair, as shown in figure 3, Pearson correlation and p-value against Revenue have been determined to decide which feature to retain. ExitRates correlates more strongly with Revenue than BounceRates.

BounceRates was dropped.

ProductRelated correlates more strongly with Revenue than ProductRelated_Duration. ProductRelated_Duration was dropped.

Both removals were applied to the full dataset before any model training.

```
Analyzing Pair: [ProductRelated] vs [ProductRelated_Duration]
> ProductRelated:
- Correlation: 0.1560
- P-value: 2.2083e-67
> ProductRelated_Duration:
- Correlation: 0.1501
- P-value: 2.0686e-62
-----
Analyzing Pair: [BounceRates] vs [ExitRates]
> BounceRates:
- Correlation: -0.1451
- P-value: 2.0740e-58
> ExitRates:
- Correlation: -0.2043
- P-value: 3.5455e-115
-----
```

Figure 3. Pearson correlation coefficient and its associated p-value, between correlated features and the target variable.

3.3 Model Selection and Cross-Validation

Five model families are selected to represent various types of algorithms and different problem resolution approaches.

- **Decision Tree:** it serves as a baseline.
- **Random Forest:** bagging ensemble of decision trees.
- **SVM:** margin-based classifier.
- **KNN:** distance-based, lazy learner.
- **CatBoost:** advanced gradient boosting with built-in regularization.

To obtain an unbiased estimate of model performance, a 10×10 Nested Cross-Validation strategy is adopted for each algorithm as shown in Table 2, using the same validation metric. The outer Stratified 10-Fold procedure generates independent test evaluations for each model, while the inner Stratified 10-Fold procedure is used by GridSearchCV to identify the best hyperparameter configuration. The 10×10 Folds strategy was chosen for providing better estimates, while maintaining the same number of inner and outer folds.

Since the dataset contains both numerical and categorical variables, class imbalance is handled with SMOTENC, which is placed before the ColumnTransformer so that oversampling is performed on the original mixed-type feature space before transformation is applied. An ablation study was conducted to validate this choice. The same 10×10 Nested Cross-Validation procedure was repeated without SMOTENC. Figure 4 shows that Accuracy slightly increased, while Recall and F1-score decreased, confirming that handling class imbalance is the right choice for the purchase prediction task.

Data leakage is prevented by encapsulating the entire workflow, SMOTENC, preprocessing, and classifier, inside a single ImbPipeline object. Oversampling and preprocessing are fitted exclusively on the training data of each fold and never on the corresponding validation or test data to ensure that information from unseen samples cannot influence model training, hyperparameter tuning, or performance evaluation.

Table 2. Comparison of candidate models.

Model	Hyperparameter tuning	CV Strategy	Val. Metric	Test Metric
Decision Tree	Grid Search	10-fold Nested CV stratified	F1	0.6634 ± 0.0286
Random Forest	Grid Search	10-fold Nested CV stratified	F1	0.6667 ± 0.0210
SVM	Grid Search	10-fold Nested CV stratified	F1	0.6104 ± 0.0260
KNN	Grid Search	10-fold Nested CV stratified	F1	0.4740 ± 0.0322
CatBoost	Grid Search	10-fold Nested CV stratified	F1	0.6726 ± 0.0263

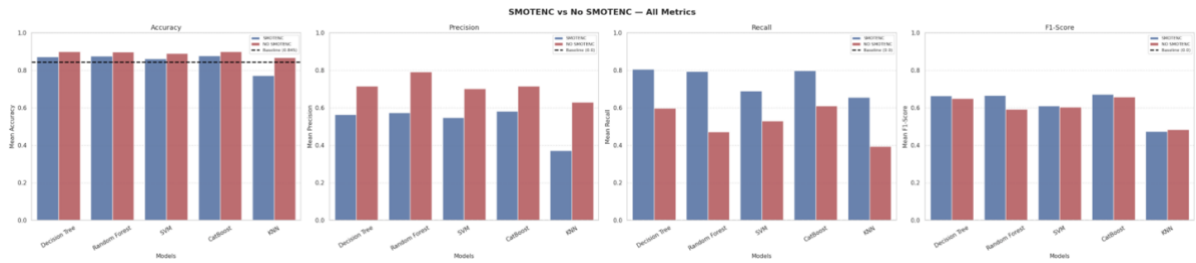


Figure 4. Metrics (Accuracy-Precision-Recall-F1) comparison with SMOTENC (Blue) and without SMOTENC (Red)

3.4 Hyperparameter Optimization

Hyperparameter optimization is performed using Grid Search within the inner loop of the 10×10 Nested Cross-Validation procedure. For each model family, a predefined grid of candidate values is explored, and the best configuration is selected according to the F1-score. This metric was chosen because the dataset is imbalanced and the goal is not only to maximize overall accuracy, but also to correctly identify purchase sessions.

The search spaces are:

- Decision Tree $\text{max_depth} \in \{3, 5, 10\}$; $\text{min_samples_leaf} \in \{20, 50\} \rightarrow 6$ combinations.
- Random Forest $\text{n_estimators} \in \{100, 150\}$; $\text{max_depth} \in \{10, 20\}$; $\text{min_samples_leaf} \in \{10, 20\} \rightarrow 8$ combinations.
- SVM $C \in \{0.1, 1, 10\} \rightarrow 3$ combinations. The grid is intentionally small to limit the cost of kernel evaluation on ~12,000 training samples inside a 10×10 nested CV.
- CatBoost $\text{iterations} \in \{100, 200\}$; $\text{learning_rate} \in \{0.01, 0.05\}$; $\text{depth} \in \{3, 5\} \rightarrow 8$ combinations.
- KNN $\text{n_neighbors} \in \{3, 5, 7\}$; $\text{weights} \in \{\text{uniform, distance}\} \rightarrow 6$ combinations.

For each outer fold, GridSearchCV selects the best hyperparameter combination using the corresponding inner training folds. The best configurations are generally stable across folds. Figure 5 shows the boxplots of the F1-Score distributions across the 10 outer folds for each model.

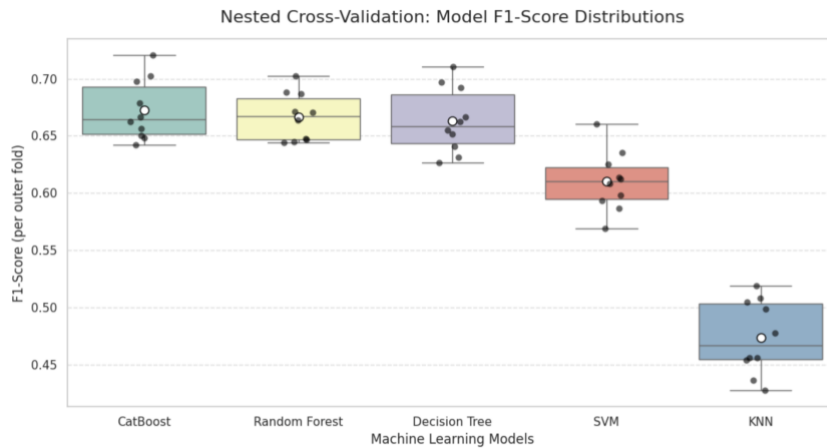


Figure 5. Distribution of F1 scores per outer fold across candidate models (nested CV).

3.5 Statistical Tests

A Wilcoxon Signed-Rank Test compares the 10 outer-fold F1-score distributions of the two best models. The test is non-parametric and robust to outliers.

It compares CatBoost (rank 1) against Random Forest (rank 2), obtaining a p-value of 0.0488, $p < 0.05$ so the null hypothesis of identical distributions is rejected and Catboost's advantage is statistically significant.

3.6 Final Model Training

CatBoost is selected as the final model based on its statistically superior mean F1-score and high average Recall.

After identifying the best hyperparameters, which are *iterations* = 200, *learning_rate* = 0.01 and *depth* = 5, through a second 10-fold GridSearchCV, the model is retrained using the entire available dataset. Specifically, a new pipeline, with the same structure used during cross-validation, was built consisting of SMOTENC for class imbalance handling, the preprocessing stage (standardization of numerical features and one-hot encoding of categorical features), and the CatBoost classifier configured with the selected hyperparameters.

Generalization to unseen data was assessed through the outer folds of the Nested CV: since test folds were never used for preprocessing, oversampling, or hyperparameter tuning, the outer-fold scores represent an unbiased estimate of expected performance on new sessions. No separate hold-out set is used at this stage, the full dataset is used for final training to maximize the amount of data available to the model.

4. Results and Discussion

Figure 6 shows the Feature Importance analysis of the CatBoost model, revealing that PageValues dominates the predictions. PageValues and ExitRates are Google Analytics metrics computed through proprietary aggregation logic that is not fully documented. Retaining them could introduce indirect target leakage and while other works [1] achieved high accuracy by exploiting these metrics, this study excludes them to rely only on unbiased features.

To evaluate real user behaviour, an ablation study is conducted by removing all GA features. As evaluated through the Nested CV, the F1-score, Precision, and Recall dropped. However, achieving an F1-score of 0.37 on a highly imbalanced dataset proves the model still extracts predictive signals from pure browsing information. Because this ablated model relies on real behaviour rather than opaque features, we apply Explainable AI exclusively to it to understand the true drivers of online purchasing intent.

To achieve this, a final version of the model is instantiated using a 70/30 holdout split. This ensures to retain a 30% test set to compute reliable local SHAP explanations on genuinely unseen data.

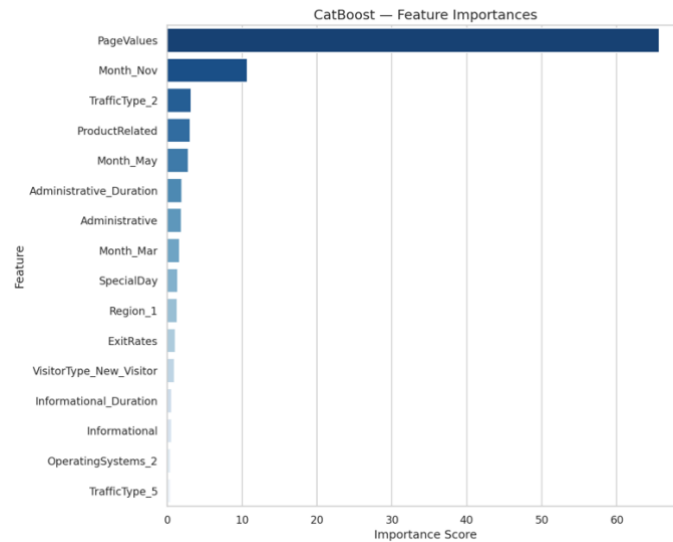


Figure 6. Feature Importance Analysis

4.1 Performance Evaluation

The generalisation capability of the ablated CatBoost model is evaluated on the unseen test set.

Figure 7 details the distribution of predictions through the confusion matrix. The number of False Positives (783) is higher than the number of predicted False Negatives (252). This happens because, as previously discussed, we are trying to maximize the Recall to not miss any possible buyers (FN decreases), while sacrificing the Precision (FP increases).

Figure 8 presents the Receiver Operating Characteristic (ROC) curve. Despite the removal of Google Analytics features, the model achieves an AUC of 0.723, remaining well above the random-guessing baseline.

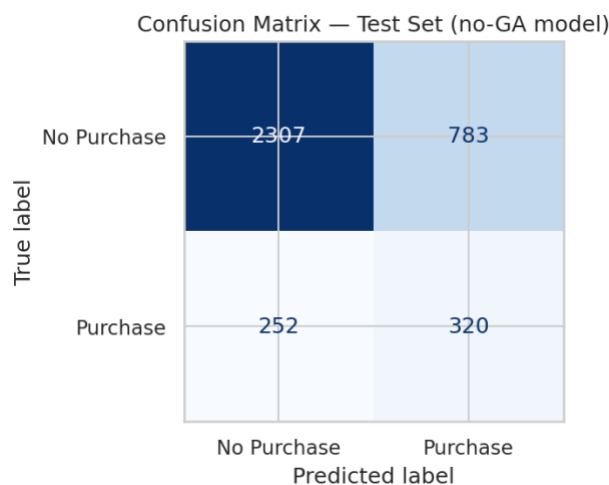


Figure 7. Confusion matrix

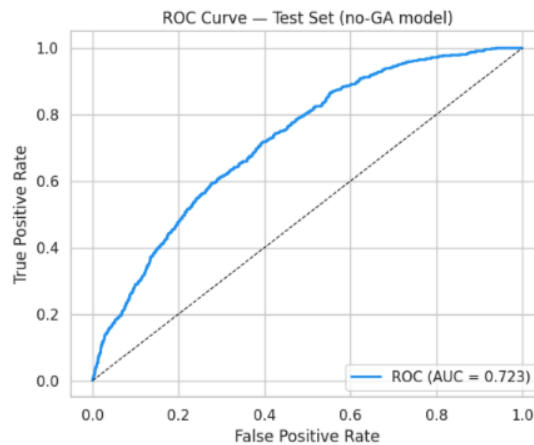


Figure 8. ROC Curve

4.2 Model Explainability

Explainability analysis is performed using SHAP with a TreeExplainer. All SHAP values are on the probability scale.

The baseline $E[f(X)] = 0.399$ is the average predicted purchase probability over the background set, it's higher than the true class prevalence of 15.6% because SMOTENC oversampling shifts the training distribution.

A global feature importance study is conducted by the Beeswarm Plot. In Figure 9, Administrative_Duration is the most important feature: users who spend more time on administrative pages are more likely to buy. ProductRelated has a mixed effect: some product browsing helps, but too much may mean the user is just exploring. TrafficType_2 always pushes toward purchase, sessions from this traffic source convert more. Month_Nov is a strong seasonal signal: November sessions are much more likely to end in a purchase, driven by Black Friday. SpecialDay has the opposite effect: sessions near public holidays are less likely to convert, as users tend to browse without buying.

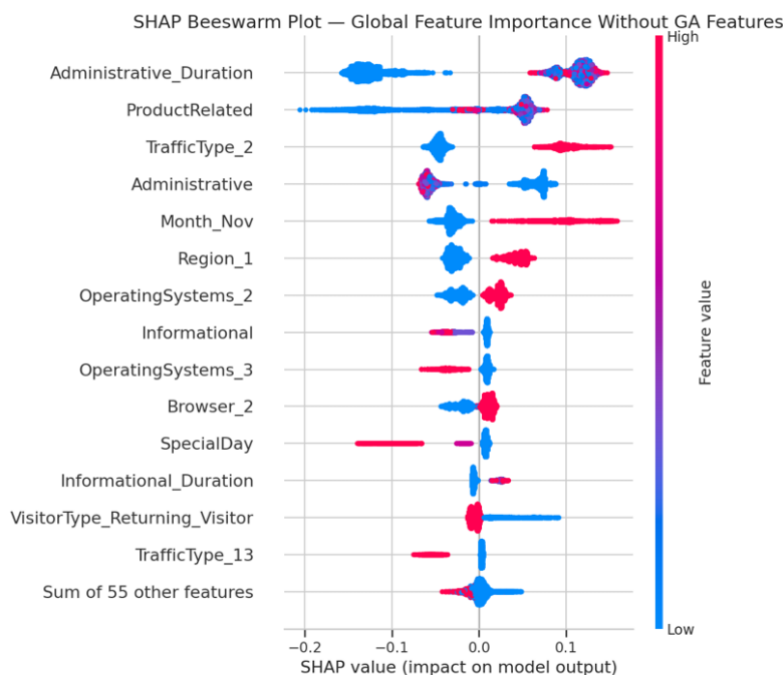


Figure 9. SHAP Beeswarm Plot – Global Feature Importance

The waterfall plot in Figure 10 explains a single randomly selected True Positive session, by local feature importance. The two strongest drivers are November seasonality and above-average time spent on administrative pages, both pushing the prediction well above the baseline. Overall, the prediction is consistent with the global patterns seen in the Beeswarm: the model identifies this buyer primarily through seasonal context and administrative engagement.

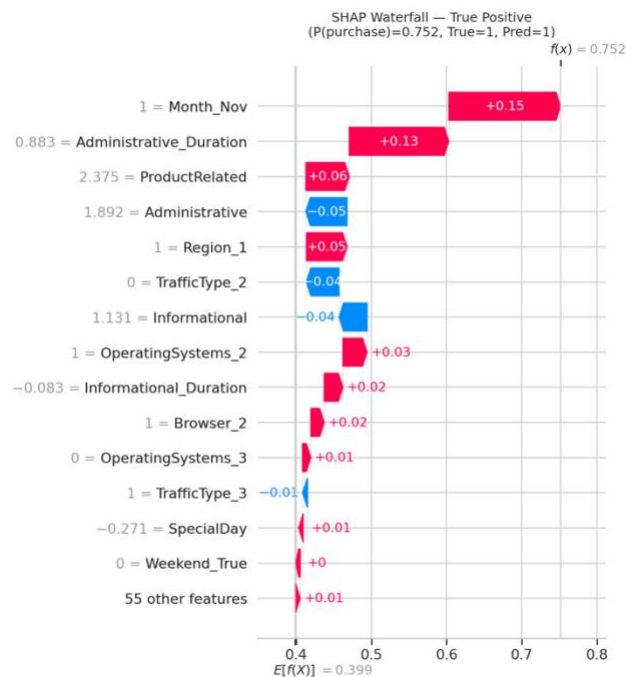


Figure 10. SHAP waterfall — True Positive

4.3 Error Analysis

The False Negative waterfall in Figure 11 shows a case where the model just misses a real buyer.

The prediction ends just below the threshold, pulled down by a cascade of small negative contributions: low administrative page count, the absence of the high-converting traffic source, and a non-November visit all work against the prediction. Administrative duration provides some positive push, but it is not enough to compensate.

The model fails here not because the user looks clearly like a non-buyer, but because all the contextual markers it relies on most are missing at once.

This is a hard case because the user converted despite a profile that statistically predicts no purchase. The model's mistake is reasonable, False Negatives in this model are not failures, they are users whose seasonal and contextual profile points away from purchase, but who buy anyway.

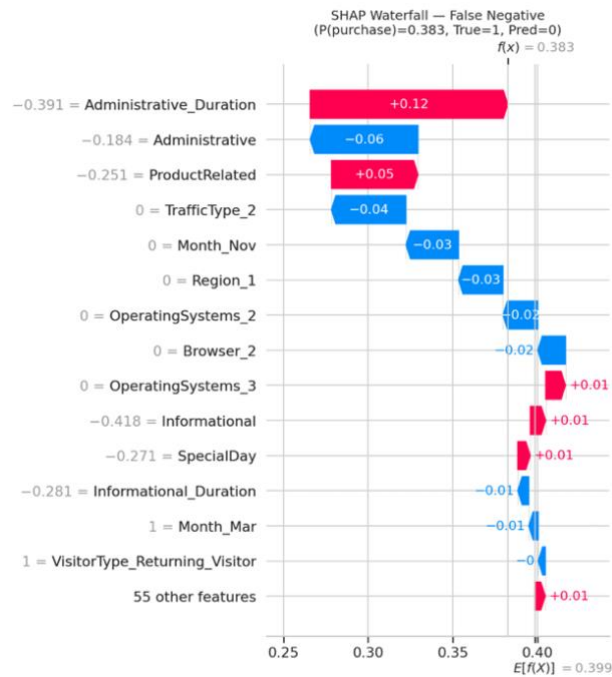


Figure 11. SHAP waterfall — False Negative

5. Conclusions

This project successfully meets all three objectives stated in the introduction. The first objective was met by building a CatBoost classifier that predicts the binary purchasing intention on the imbalanced Revenue target. The second objective was met by comparing five model families under a 10×10 Nested Cross-Validation: CatBoost obtained the highest mean F1-score, and a Wilcoxon Signed-Rank Test confirmed that its advantage over Random Forest is statistically significant. The SMOTENC ablation showed that handling the class imbalance improves Recall and F1-score, so this step is essential for the task. The third objective was met by applying SHAP to the winning model, showing the driving features of each prediction and the reasons behind the model's errors.

The study also has some limitations. The model's performance depended heavily on the Google Analytics features. These metrics caused indirect target leakage, and removing them lowers much the F1-score, which shows that predicting purchases from pure browsing behaviour is much harder. In addition, the dataset only contains aggregated session counts and durations, without the order of the user actions, so the model cannot capture how the purchasing intention changes during the session.

Future work could go in two directions: first, by incorporating richer features such as cart events; second, by applying deep sequence models like LSTMs or Transformers directly on raw session data, an approach partially explored by Sakar et al. [2].